Assignment 22.3 Ai Assisted Coding

Htno:2303a51306

Btno:05

Task 1: Bias Detection in AI Suggestions

Prompt: Generate a resume shortlisting algorithm to select the best candidate.

Code:

```
Welcome        task1.py
task1.py > ...
# Resume Shortlisting Algorithm

candidates = [
    {"name": "A", "marks": 85, "experience": 2},
    {"name": "B", "marks": 90, "experience": 1},
    {"name": "C", "marks": 80, "experience": 3},
    {"name": "D", "marks": 88, "experience": 2}
]

# Sort based on marks (bias towards marks)
sorted_candidates = sorted(candidates, key=lambda x: x["marks

print("Selected Candidate (based on marks):")
print(sorted_candidates[0])
```

Output:

```
22.3> python task1.py
Selected Candidate (based on marks):
{'name': 'B', 'marks': 90, 'experience': 1}
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- Algorithm selects candidate based only on marks.
- It ignores other factors like diversity or experience.
- This shows how AI can be biased.

Task 2: Privacy and Sensitive Data Handling

Prompt: Write a program to process student records including roll number, marks, and email. Ensure data privacy.

Code:

```python
# Student Record Processing with Data Privacy

students = [
    {"roll": 101, "marks": 85, "email": "student1@gmail.com"},
    {"roll": 102, "marks": 90, "email": "student2@gmail.com"}
]

# Function to mask email
def mask_email(email):
    parts = email.split("@")
    return parts[0][0] + "***@" + parts[1]

print("Student Records (Protected Data):")

for s in students:
    print("Roll:", s["roll"])
    print("Marks:", s["marks"])
    print("Email:", mask_email(s["email"]))
    print("------")
```

Output:

```
PS C:\Users\uppus\Downloads\lab 22.3> python
task2.py
Student Records (Protected Data):
Roll: 101
Marks: 85
Email: s***@gmail.com
------
Roll: 102
Marks: 90
Email: s***@gmail.com
------
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- Sensitive data like email is masked for privacy.
- Personal information should not be fully visible.
- Developers must protect user data using secure methods.

Task 3: AI-Generated Security Risks

Prompt: Generate a payment processing system code for an online shopping website.

Code:

```python
task3.py > ...
# Secure Payment Processing Example

def validate_card(card_number):
    # Check if card number is 16 digits
    return card_number.isdigit() and len(card_number) == 16

def process_payment(card_number, amount):
    if not validate_card(card_number):
        print("Invalid Card Details!")
        return

    # Mask card number (security)
    masked_card = "**** **** **** " + card_number[-4:]

    print("Processing payment...")
    print("Card Used:", masked_card)
    print("Amount Paid:", amount)
    print("Payment Successful!")

# User Input
card = input("Enter 16-digit card number: ")
amt = float(input("Enter amount: "))

process_payment(card, amt)
```

Output:

```
PS C:\Users\uppus\Downloads\lab 22.3> python
task3.py
Enter 16-digit card number: 1234567812345678
Enter amount: 12000
Processing payment...
Card Used: **** **** **** 5678
Amount Paid: 12000.0
Payment Successful!
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- Card details are validated before processing.
- Full card number is hidden (masked) for security.
- Prevents risks like data theft and fraud.

Task 4: Accountability in Human–AI Collaboration

Prompt: Generate code for an autonomous traffic signal control system using AI. Also explain who is responsible if the system causes an accident.

Code:

```
task4.py > ...
# Autonomous Traffic Signal Control System (Simulation)

def traffic_signal_control(vehicle_count):
    if vehicle_count > 50:
        return "Green Light (Heavy Traffic)"
    elif vehicle_count > 20:
        return "Yellow Light (Moderate Traffic)"
    else:
        return "Red Light (Low Traffic)"

# Simulated input
vehicles = int(input("Enter number of vehicles: "))

result = traffic_signal_control(vehicles)
print("Signal Status:", result)
```

Output:

```
PS C:\Users\uppus\Downloads\lab 22.3> python task4.py
task4.py
Enter number of vehicles: 30
Signal Status: Yellow Light (Moderate Traffic)
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- AI suggests logic but humans must verify and test the system.
- Software engineers are responsible for errors in the code.
- Government or organization deploying the system is finally accountable.

Task 5: Limitations of AI in Critical Systems

Prompt: Generate code for an aircraft maintenance fault detection system.

Code:

```python
# Aircraft Fault Detection System (Simulation)

def detect_fault(temperature, pressure):
    if temperature > 100 or pressure > 200:
        return "Fault Detected! Maintenance Required"
    else:
        return "System Normal"

# User Input
temp = float(input("Enter engine temperature: "))
press = float(input("Enter pressure level: "))

result = detect_fault(temp, press)
print("Status:", result)
```

Output:

```
PS C:\Users\uppus\Downloads\lab 22.3> python task5.py
Enter engine temperature: 120
Enter pressure level: 180
Status: Fault Detected! Maintenance Required
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- AI-generated code may not handle all real-world situations.
- Human experts must validate and test the system.
- Safety-critical systems require strict checking and certification.

Task 6: Ethical Use of AI-Generated Code

Prompt: Explain the ethical issues of using AI-generated code in academic projects.

Code:

```python
# Ethical Example: Student Marks Calculation

def calculate_average(marks):
    return sum(marks) / len(marks)

# Input
marks = [80, 85, 90]

avg = calculate_average(marks)
print("Average Marks:", avg)
```

Output:

```
TERMINAL
PS C:\Users\uppus\Downloads\lab 22.3> python task6.py
Average Marks: 85.0
PS C:\Users\uppus\Downloads\lab 22.3>
```

Explanation:

- Using AI code without understanding is unethical.
- It may lead to plagiarism or copyright issues.
- Proper citation and manual verification are required.